

ZIHAN LIN

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EDUCATION

Georgia Institute of Technology

B.S. Biomedical Engineering | Minor in Robotics

Atlanta, GA

Expected May 2028

University of Florida

B.S. Biomedical Engineering | Dean's List, Herbert Wertheim College of Engineering

Gainesville, FL

Aug 2024 - Dec 2025

ROBOTICS EXPERIENCE

Research Contributor - Soft Robotics Berry Harvesting Project

Atlanta, GA

BioMedical Mechatronics Lab (BM2) | Georgia Institute of Technology

Jan 2026 - Present

- Developed a MuJoCo simulation pipeline for a berry-picking manipulator by converting and organizing CAD and URDF assets from an existing hardware platform.
- Processed 100+ STL components and resolved coordinate-frame, mesh-orientation, origin, and link-hierarchy issues during robot model import.
- Structured the environment for upcoming inverse kinematics and motion-planning experiments, with planned ROS 2 integration, computer-vision-based fruit detection and pose estimation, visual servoing, and learning-based manipulation.

Custom 6-DOF Robotic Arm

Shanghai, China

Independent Project | Python, STM32, MuJoCo, Trajectory Planning, CAD

Jun 2026 - Present

- Designed and assembled a custom 6-DOF arm with 3D-printed joints, timing-belt reductions, bearings, shafts, and modular actuator mounts.
- Implemented STM32-based stepper motor control and serial communication with a Python desktop GUI for joint-level commands and rapid hardware debugging.
- Evaluated pulley ratios, joint torque, belt tension, backlash, shaft support, and structural stiffness to guide mechanical iteration.
- Built a MuJoCo digital twin and Python trajectory-planning pipeline that enables the simulated end effector to reproduce handwriting paths; extending the system toward encoder-based closed-loop execution, visual servoing, and learning-based manipulation.

Independent Project - MRI-Compatible Robotic Needle Intervention Platform

Atlanta, GA

BioMedical Mechatronics Lab (BM2) | Georgia Institute of Technology | Guided by Samuel Wilcox

Jan 2026 - Present

- Designed and prototyped a modular MRI-compatible robotic needle intervention platform for precision image-guided minimally invasive procedures.
- Reverse-engineered legacy pneumatic actuator concepts and redesigned mechanical subsystems to improve manufacturability, modularity, and integration with MR-safe robotic hardware.
- Created multi-DOF needle-positioning mechanisms in SolidWorks, including actuator linkages, ball-joint interfaces, pivot constraints, worm-drive concepts, and modular needle-insertion assemblies, with interfaces reserved for future image guidance and closed-loop control.

TECHNICAL SKILLS

Programming: Python, C, MATLAB

Robotics & Simulation: MuJoCo, URDF/MJCF, STM32, serial communication, robot kinematics, sensor-actuator integration

Engineering & Hardware: SolidWorks, CAD modeling, Arduino, 3D printing, stepper motors, mechanical prototyping